

Francesco Bianchi

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EDUCATION

University of Trento (UNITN)

Master of Science in Computer Science (Cybersecurity track)

Sep 2025 – Present

Trento, Italy

University of Milan-Bicocca

Bachelor of Science in Computer Science – Final Grade: 104/110

Oct 2022 – Jul 2025

Milan, Italy

High School: Liceo delle Scienze Applicate “Sottotenente di Vascello Antonio Badoni”
2022

Sep 2017 – Jul

Applied Sciences specialization

Lecco, Italy

EXPERIENCE

Internship – Research Thesis Project

University of Milan-Bicocca

Mar 2025 – Jul 2026

Milan, Italy

- Designed and implemented an **automatic evaluation system for LLM responses**, classifying outputs as jailbreak or non-jailbreak.
- Fine-tuned a **BERT-based classifier** on a curated dataset of 1,784 labeled instances, improving robustness and accuracy over baseline models.
- Integrated **unsupervised clustering (K-means)** to explore semantic patterns and latent structures in LLM responses.
- Built a modular and containerized software architecture in Python, ensuring reproducibility and portability.
- Conducted extensive validation with visualization techniques (PCA, t-SNE, UMAP) to analyze clustering quality and semantic separation.
- Contributed to a broader AI security research project at the Department of Informatics (DISCo), focusing on alignment and jailbreak resilience.

Assistant Technician (Part-time)

Pitek

Jun 2021 – Oct 2021

Lecco, On-site

- Assisted with IT infrastructure management for client companies.
- Monitored and maintained servers and network systems to ensure proper functionality.
- Provided on-site support for customers, troubleshooting issues and ensuring service continuity.

PROJECTS

Portfolio Website

Personal Project

Jan 2025 – Present

Milan, Remote

- Built a responsive portfolio website using Next.js, TypeScript, and Framer Motion.
- Designed with a minimalist, Swiss-inspired aesthetic emphasizing typography and smooth animations.
- Optimized loading times and implemented SEO best practices for better visibility.
- Developed mobile-first responsive layouts to ensure cross-device compatibility.

SmartHome IoT Simulator

University of Trento (Service Design Course)

Oct 2025 – Jan 2026

Trento, On-site

- Co-designed and developed a smart home IoT simulator based on a microservices (SOA) architecture.
- Built the backend using Python 3 and Flask, exposing RESTful APIs for seamless cross-layer communication.
- Containerized and orchestrated the multi-container environment using Docker and Docker Compose to ensure scalability and isolation.
- Implemented real-time anomaly detection algorithms (e.g., smoke alerts) and energy efficiency optimization logic.
- Integrated MongoDB for robust data persistence and interfaced with external APIs (Telegram Bot, OpenWeatherMap) to power a monitoring Web Dashboard.

Ignition Finance Android App

University of Milan-Bicocca

Nov 2024 – Feb 2025

Milan, Hybrid

- Built an Android app using Kotlin and Jetpack Compose, focusing on performance and usability.
- Developed a simulation engine capable of processing 1,000 runs of 100 years of market data in 7 seconds.
- Implemented interactive visualizations for portfolio analysis with dynamic graphs and charts.

- Integrated Firebase for authentication and data persistence.
- Worked in a multidisciplinary team to deliver a scalable and well-tested product.
- Awarded top marks by the supervising professor.

Domains Web App

University of Milan-Bicocca

Jun 2024

Milan, Hybrid

- Designed and developed a full-stack platform for domain registration, renewal, and reacquisition.
- Implemented REST APIs with Java, Maven, Jetty, and Jersey to manage transactions and users.
- Built a document-based database using HashMap collections for efficient data handling.
- Created a responsive front-end with HTML, CSS, and JavaScript from Figma designs.
- Ensured smooth client-server interaction with a custom CORS filter.
- Delivered the solution in collaboration with a multidisciplinary team under tight deadlines.

Prolog and Lisp Object-Oriented Extensions

University of Milan-Bicocca

Nov 2023 – Jan 2024

Milan, Remote

- Extended Prolog and Lisp with OOP features such as inheritance, fields, and methods.
- Ensured type consistency for reliable data management.
- Completed within strict deadlines, showcasing strong time management and teamwork.

Dsh.C – A Simple Shell in C

Personal Project

Dec 2023 – Jan 2024

Milan, Remote

- Implemented a Unix shell in C supporting path management, redirection, and piping.
- Executed commands via search paths and allowed dynamic path setting.
- Supported output redirection (`<` and `<<`) and piping (`|`).
- Designed as an educational project using `sys/wait` in a Linux environment.

SKILLS

Programming Languages: Java, Kotlin, Python, Julia, C, SQL, JavaScript, Prolog, Lisp

Frameworks & Technologies: Flask, Next.js, TypeScript, Jetpack Compose, REST APIs, Docker, Docker Compose, Firebase, MongoDB, Git

AI & Machine Learning: LLM evaluation, fine-tuning (BERT), unsupervised clustering (K-means), PCA/t-SNE/UMAP, jailbreak techniques

UI/UX & Tools: Figma, Adobe InDesign, Design Systems

Languages: Italian (Native), English (C1)